THE TOP COKERNEL OF A BOUNDED ABOVE FLAT COMPLEX OF FINITE TOR AMPLITUDE IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- Tor amplitude of a complex and the definition of the interval [a, b]. defined earlier in the same section, source lines 17100-17236 is later; the notion is used here from the preceding block around 16800-16881. Supplies the hypothesis on the complex.
- A bounded above complex of flat modules computes the derived tensor product, so the ordinary tensor product has the same homology. used at proof lines 16889-16892. Justifies using the ordinary complex to compute Tor.
- Flatness of a module tested through the vanishing of Tor_1 against every module. the author cites Algebra, Lemma algebra-lemma-characterize-flat at proof lines 16899-16902. Converts the vanishing statement into flatness.

Source reference. Original lemma 0653: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7887–7887. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00M5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 811–811. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $K^\bullet$ be a bounded above complex of flat R-modules with tor-amplitude in $[a, b]$. Then $\mathrm{Coker}(d_K^{a-1})$ is a flat R-module.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. As $K^\bullet$ is a bounded above complex of flat modules we see that $K^\bullet \otimes_R M = K^\bullet \otimes_R^{\mathbf{L}} M$. Hence for every R-module M the sequence

$$K^{a-2} \otimes_R M \to K^{a-1} \otimes_R M \to K^a \otimes_R M$$

is exact in the middle. Since $K^{a-2} \to K^{a-1} \to K^a \to \mathrm{Coker}(d_K^{a-1}) \to 0$ is a flat resolution this implies that $\mathrm{Tor}_1^R(\mathrm{Coker}(d_K^{a-1}), M) = 0$ for all R-modules M. This means that $\mathrm{Coker}(d_K^{a-1})$ is flat, see Algebra, Lemma 00M5. $\square$